

# MSC2050 Discrete Mathematics, week 12

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Version 1.0 typed under  $\text{\LaTeX}$

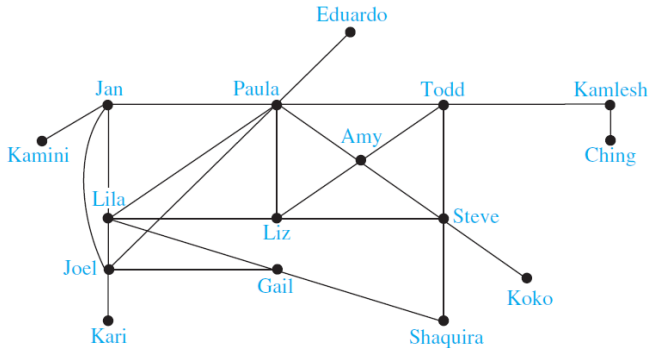
# *Graphs and Graph models*

*(Paragraph 10.1)*

A graph  $G = (V, E)$  consists of  $V$ , a nonempty set of **vertices** (or nodes) and  $E$ , a set of **edges**.

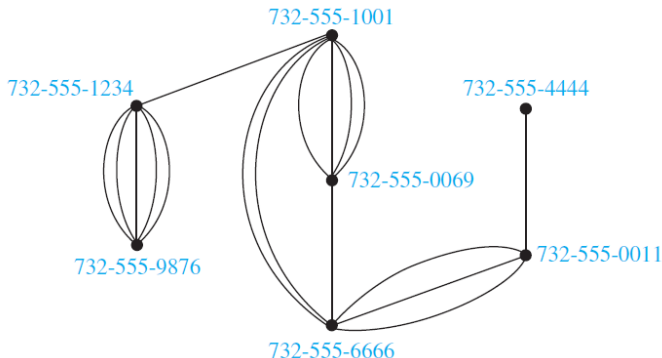
Each edge has either one or two vertices associated with it, called its **endpoints**.

An edge is said to **connect** its endpoints.



A graph in which each edge connects two different vertices and where no two edges connect the same pair of vertices is called a **simple graph**.

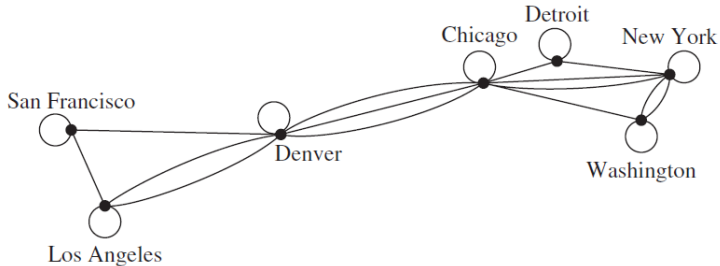
Graphs that may have multiple edges connecting the same vertices are called **multigraphs**.



Here graph models telephone calls. When we care only whether there has been a call connecting two telephone numbers, An edge connects telephone numbers when there has been a call between these numbers.

Edges that connect a vertex to itself are called **loops**, and sometimes we may even have more than one loop at a vertex.

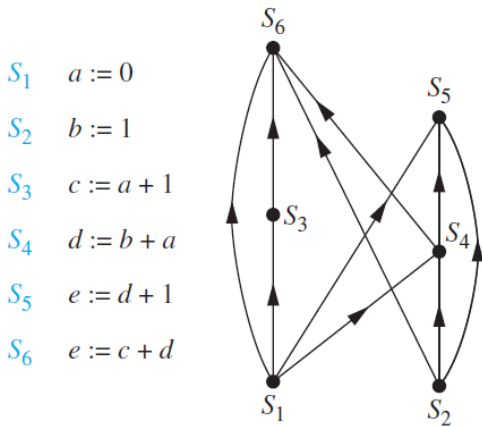
Graphs that may include loops, and possibly multiple edges connecting the same pair of vertices or a vertex to itself, are called **pseudographs**.



Sometimes a communications link connects a data center with itself, perhaps a feedback loop for diagnostic purposes.

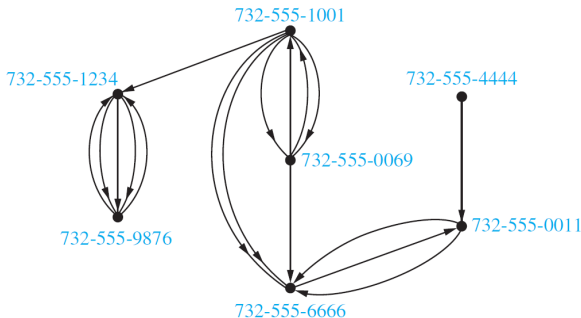
In a **directed graph** (or digraph) each directed edge is associated with an ordered pair of vertices.

The directed edge associated with the ordered pair  $(u, v)$  is said to start at  $u$  and end at  $v$ .



A Precedence Graph

Directed graphs that may have multiple directed edges from a vertex to a second (possibly the same) vertex are called **directed multigraphs**. When there are  $m$  directed edges, each associated to an ordered pair of vertices  $(u, v)$ , we say that  $(u, v)$  is an edge of **multiplicity**  $m$ .



The edge representing a call starts at the telephone number from which the call was made and ends at the telephone number to which the call was made. For example,  $(732-555-1001, 732-555-6666)$  has multiplicity 2.

# Graph terminology

| <i>Type</i>           | <i>Edges</i>            | <i>Multiple Edges Allowed?</i> | <i>Loops Allowed?</i> |
|-----------------------|-------------------------|--------------------------------|-----------------------|
| Simple graph          | Undirected              | No                             | No                    |
| Multigraph            | Undirected              | Yes                            | No                    |
| Pseudograph           | Undirected              | Yes                            | Yes                   |
| Simple directed graph | Directed                | No                             | No                    |
| Directed multigraph   | Directed                | Yes                            | Yes                   |
| Mixed graph           | Directed and undirected | Yes                            | Yes                   |



### Homework:

- Solve # 1,3,5,7,9,19,21,27,29,33 at p. 649-651.

Answer Q1

# *Special types of graphs and graph isomorphism*

*(Paragraphs 10.2 and 10.3)*

# Undirected graph

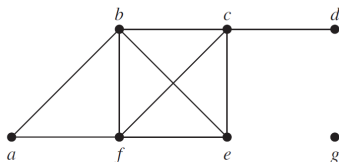
Two vertices  $u$  and  $v$  in an undirected graph  $G$  are called **adjacent** (or **neighbors**) in  $G$  if  $u$  and  $v$  are endpoints of an edge  $e$  of  $G$ .

Such an edge  $e$  is called **incident** with the vertices  $u$  and  $v$ .

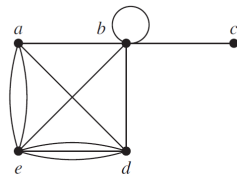
The set of all neighbors  $v$  denoted by  $N(v)$ , is called the **neighborhood** of  $v$ .

The **degree** of a vertex is the number of edges incident with it, except that a loop at a vertex contributes twice to the degree of that vertex.

The degree of the vertex  $v$  is denoted by  $\deg(v)$ .



$G$



$H$

In  $G$ :  $\deg(a) = 2$ ,  $\deg(d) = 1$ ,  $\deg(f) = 4$ ,  $\deg(g) = 0$ , (isolated point)

$N(a) = \{b, f\}$ ,  $N(d) = \{c\}$ ,  $N(f) = \{a, b, c, e\}$ ,  $N(g) = \emptyset$ .

In  $H$ :  $\deg(b) = 6$ ,  $\deg(e) = 6$ ,  $N(b) = \{a, b, c, d, e\}$ ,  $N(e) = \{a, b, d\}$ .

# THE HANDSHAKING THEOREM

Let  $G = (V, E)$  be an undirected graph with  $m$  edges. Then

$$2m = \sum_{v \in V} \deg(v).$$

(Note that this applies even if multiple edges and loops are present.)

## Theorem

An undirected graph has an even number of vertices of odd degree.

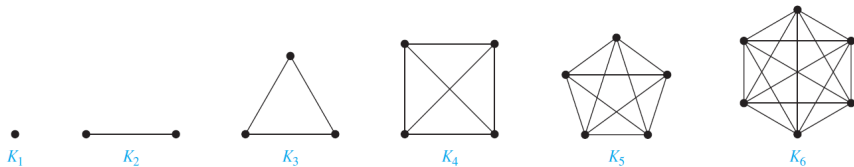
proof: Let  $V_1$  and  $V_2$  be the set of vertices of even degree and the set of vertices of odd degree, respectively. Then

$$2m = \sum_{v \in V} \deg(v) = \sum_{v \in V_1} \deg(v) + \sum_{v \in V_2} \deg(v).$$

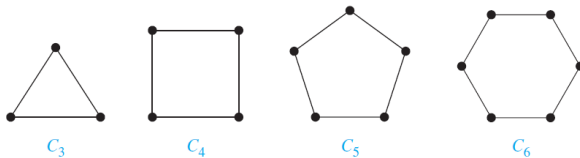
$2m$  and  $\sum_{v \in V_1} \deg(v)$  are even, so  $\sum_{v \in V_2} \deg(v)$  must be even.

# Some special simple graphs

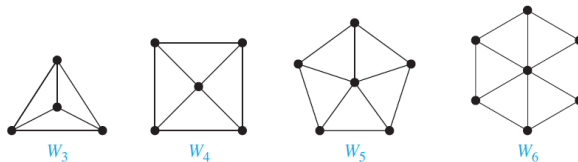
## Complete graphs



## Cycles

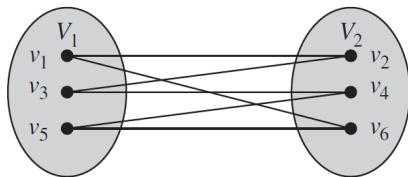


## Wheels



# Bipartite graphs

A simple graph  $G$  is called bipartite if its vertex set  $V$  can be partitioned into two disjoint sets  $V_1$  and  $V_2$  such that every edge in the graph connects a vertex in  $V_1$  and a vertex in  $V_2$ .



For example  $C_6$  is bipartite.

Any complete graph is not bipartite.

A simple graph is bipartite if and only if it is possible to assign one of two different colors to each vertex of the graph so that no two adjacent vertices are assigned the same color.

Answer Q2

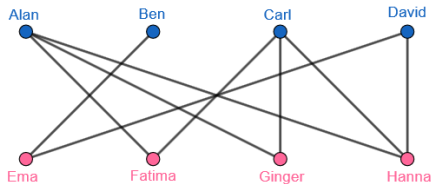
# Bipartite graphs and matching

Graph matching problems are very common in daily activities: from online matchmaking and dating sites, to medical residency placement programs.

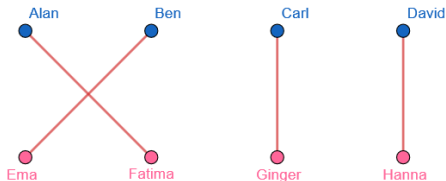
Suppose that there are  $n$  men and  $n$  women on an island.

Each person has a list of members of the opposite gender acceptable as a spouse.

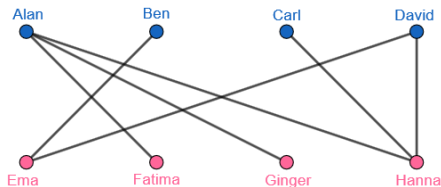
We construct a bipartite graph  $G = (V_1, V_2)$  where  $V_1$  is the set of men and  $V_2$  is the set of women so that there is an edge between a man and a woman if they find each other acceptable as a spouse.



A **complete matching** is a set of married couples where every man is married.



# Complete matching is not always possible



## HALL'S MARRIAGE THEOREM (Philip Hall in 1935)

The bipartite graph  $G = (V, E)$  with bipartition  $(V_1, V_2)$  has a complete matching from  $V_1$  to  $V_2$  if and only if  $|N(A)| \geq |A|$  for all subsets  $A$  of  $V_1$ .

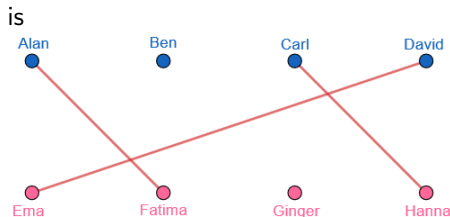
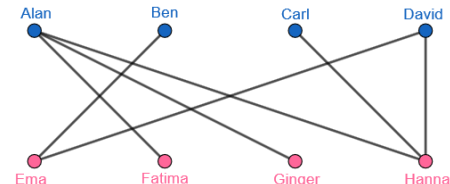
In the example above take  $A = \{\text{Ben, Carl, David}\}$ .



# Maximal matching

A **maximum matching** is a largest possible set of married couples.

A maximal matching for



## Maximal matching algorithms

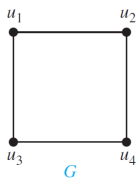
- Hungarian Maximum Matching Algorithm or Kuhn-Munkres algorithm (1957,  $O(n^3)$ )
- Blossom Algorithm (1961,  $O(|E|n^2)$ , where  $|E|$  is a number of edges)
- Hopcroft-Karp Algorithm (1973,  $O(|E|\sqrt{n})$ )

# Isomorphisms of graphs

The simple graphs  $G_1 = (V_1, E_1)$  and  $G_2 = (V_2, E_2)$  are **isomorphic** if there exists a one-to-one and onto (bijection) function  $f$  from  $V_1$  to  $V_2$  with the property that  $a$  and  $b$  are adjacent in  $G_1$  if and only if  $f(a)$  and  $f(b)$  are adjacent in  $G_2$ , for all  $a$  and  $b$  in  $V_1$ .

Such a function  $f$  is called an **isomorphism**.

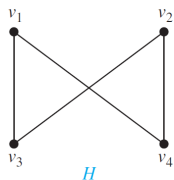
Two simple graphs that are not isomorphic are called **non-isomorphic**.



Two graphs  $G$  and  $H$  are isomorphic.

The isomorphism  $f : G \rightarrow H$  is

$f(u_1) = v_1$ ,  $f(u_2) = v_3$ ,  $f(u_3) = v_4$  and  $f(u_4) = v_2$ .

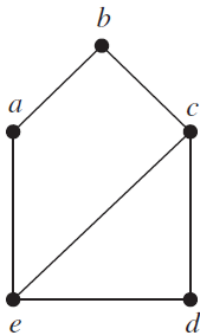


Is there other isomorphisms?

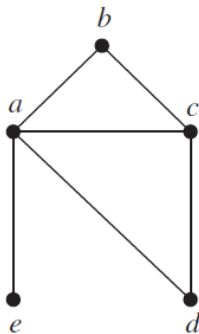
yes

For example,  $f(u_1) = v_1$ ,  $f(u_2) = v_4$ ,  $f(u_3) = v_3$  and  $f(u_4) = v_2$ .

Show that two graphs below are not isomorphic.



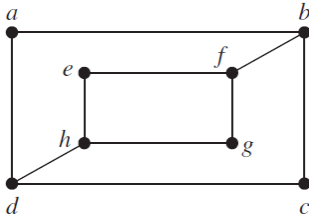
$G$



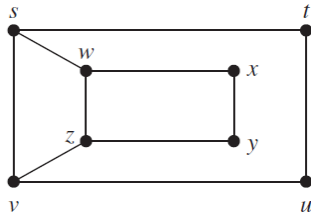
$H$

Solution: Both graphs have 5 vertices and 6 edges.  
The graph  $H$  has a vertex of degree 1, which  $G$  does not have.  
Hence,  $G$  and  $H$  are not isomorphic.

Are the following graphs isomorphic?



*G*



*H*

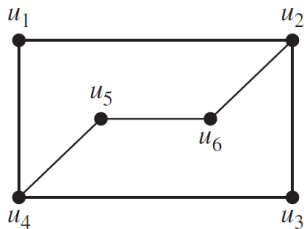
Solution: Both graphs have 8 vertices and 10 edges.

Both graphs have 4 vertices of degree 2 and 4 vertices of degree 3.

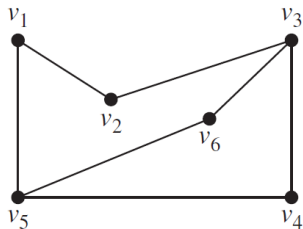
The point  $a$  of graph  $G$  is of degree 2 and connected with two vertices of degree 3.

There are no such vertex in  $H$ , so these graphs are not isomorphic.

Are the following graphs isomorphic?



*G*



*H*

Solution: Both graphs have 6 vertices and 7 edges.

Both graphs have 4 vertices of degree 2 and 2 vertices of degree 3.

Both graphs contain two subsequent points of degree 2 between the point of degree 3 and one point of degree 2 which is connected to points of degree 3.

The graphs can be isomorphic. Let's try to construct an isomorphism  $f$ .

$f(u_1) = v_6$ ,  $f(u_2) = v_3$ ,  $f(u_3) = v_4$ ,  $f(u_4) = v_5$ ,  $f(u_5) = v_1$  and  $f(u_6) = v_2$ .

So the graphs are isomorphic.

Answer Q3

## Algorithms for graph isomorphism

The best algorithms known for determining whether two graphs are isomorphic have **exponential** worst-case time complexity (in the number of vertices).

However, **linear average-case time complexity** algorithms are known that solve this problem, and there is some hope, that an algorithm with polynomial worst-case time complexity for determining whether two graphs are isomorphic can be found.

## Applications of graph isomorphism

- chemistry: use multigraphs, known as **molecular graphs**, to model chemical compounds. In these graphs, vertices represent atoms and edges represent chemical bonds between these atoms.

When a potentially new chemical compound is synthesized, a database of molecular graphs is checked to see whether the molecular graph of the compound is the same as one already known.

- design of electronic circuits: using graphs in which vertices represent components and edges represent connections between them.  
Modern integrated circuits, known as **chips**, are miniaturized electronic circuits, often with millions of transistors and connections between them.  
Graph isomorphism is the basis for the verification that a particular a circuit produced corresponds to the original schematic of the design.

### Homework:

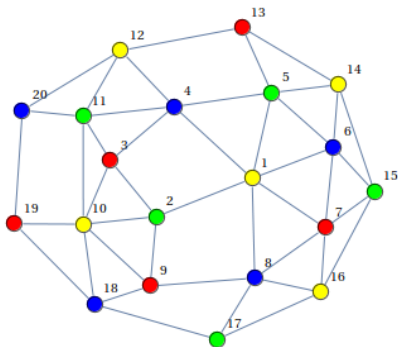
- Solve exercises # 1,3,5,7,9,11,13,21,23,25,27,29 at p. 665-666.
- Read p. 668-671 of 10.3.
- Solve exercises # 1,3,5,7,9,11,13,15,17,19,21,23,35,37,3,41,43,45,55,57 at p. 675-677.

# *Graph Coloring*

*(Paragraph 10.8)*



A **coloring** of a simple graph is the assignment of a color to each vertex of the graph so that no two adjacent vertices are assigned the same color.

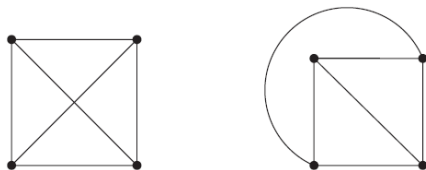


The **chromatic number**  $\chi(G)$  of a graph  $G$  is the least number of colors needed for a coloring of this graph.

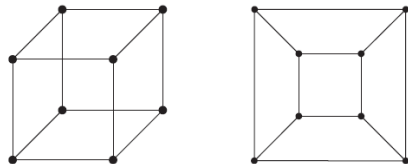
# Planar graphs (paragraph 10.7)

A graph is called **planar** if it can be drawn in the plane without any edges crossing.

The graph  $K_4$  without crossings:

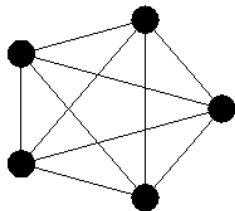


The graph  $Q_3$  without crossings:

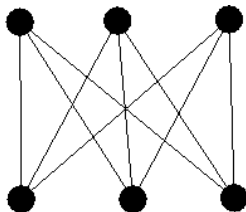


# Examples of non-planar graph

$K_5$



$K_{3,3}$

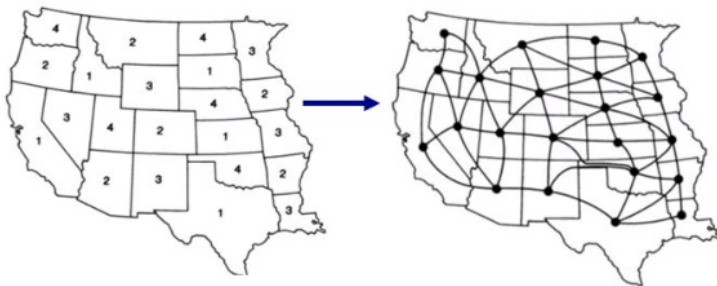


## Kuratowski's Theorem

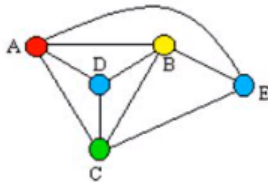
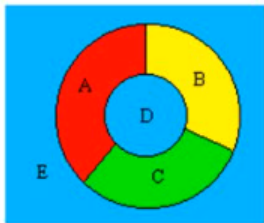
A graph is non-planar if and only if it contains a subgraph homeomorphic to  $K_{3,3}$  or  $K_5$ .

# Maps vs graphs

If there is a map we can always draw a corresponding (planar) graph.  
Only for planar graphs it is possible to construct a corresponding map.



Asking for the chromatic number of a planar graph is the same as asking for the minimum number of colors required to color a map so that no two adjacent regions are assigned the same color.



This question has been studied for more than 100 years.  
The answer is provided by one of the most famous theorems in mathematics.

## THE FOUR COLOR THEOREM

The chromatic number of a planar graph is no greater than four.

The four color theorem was originally posed as a conjecture in the 1850s. It was finally proved by the American mathematicians Kenneth Appel and Wolfgang Haken in 1976.

# Scheduling Final Exam

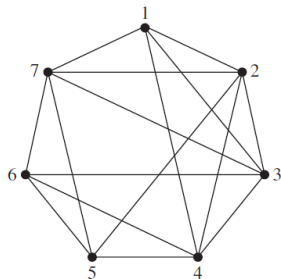
How can the final exams at a university be scheduled so that no student has two exams at the same time?

Solution: This scheduling problem can be solved using a graph model, with vertices representing courses and with an edge between two vertices if there is a common student in the courses they represent.

Each time slot for a final exam is represented by a different color.

A scheduling of the exams corresponds to a coloring of the associated graph.

For instance, suppose there are seven finals to be scheduled. Suppose the courses are numbered 1 through 7. Suppose that the following pairs of courses have common students: 1 and 2, 1 and 3, 1 and 4, 1 and 7, 2 and 3, 2 and 4, 2 and 5, 2 and 7, 3 and 4, 3 and 6, 3 and 7, 4 and 5, 4 and 6, 5 and 6, 5 and 7, and 6 and 7.



Because the chromatic number of this graph is 4, four time slots are needed. The possible answer is I-1,6, II-2, III-3,5, IV-4,7

### Homework:

- Solve exercises # 5,7,9,11,13,17,19 at p. 732-733.

Answer Q4